Fiscal Year:	FY 2014	**Task Last Updated:**	FY 12/05/2014
PI Name:	Stutte, Gary Ph.D.		
Project Title:	Symbiotic Nodulation in a Reduced Gravity Environment (SyNRGE)		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	gary.w.stutte@nasa.gov	**Fax:**	FY 321-861-2922
PI Organization Type:	NASA CENTER	**Phone:**	321-861-3493
Organization Name:	QinetiQ-North America/NASA Kennedy Space Center		
PI Address 1:	ESC-Team QNA		
PI Address 2:	Space Life Sciences Laboratory		
PI Web Page:			
City:	Kennedy Space Center	**State:**	FL
Zip Code:	32899	**Congressional District:**	24
Comments:	NOTE: PI had a change of employment from Dynamac Corporation to QinetiQ-North America, effective 1 March, 2011.		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	03/01/2011	**End Date:**	04/30/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-135 ; ISS NOTE: New end date is 4/30/2014, per NSSC information (Ed., 2/19/14) NOTE: New end date is 2/28/2014, per NSSC information (Ed., 8/6/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Roberts, Michael (CASIS-Center for the Advancement of Science in Space)		
Grant/Contract No.:	NNX11AL06A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	SyNRGE (Symbiotic Nodulation in a Reduced Gravity Environment) will investigate microgravity effects associated with microbe-host interactions and cell-cell communication using a plant-bacteria model system. Seeds from a single cultivar of the model legume species Medicago truncatula (barrel medic) will be germinated on-orbit in the presence of genetically marked strains of nitrogen-fixing bacteria of the species Sinorhizobium meliloti capable of forming a mutualistic endosymbiosis with leguminous plants. Cellular, molecular, and morphological analyses of plant tissue specimens and bacteria fixed at serial time intervals on-orbit will be compared to ground controls to elucidate the effect of microgravity on plant and bacterial physiological responses, mechanisms of cell-cell communication and resource exchange (i.e., carbon and nitrogen), and the extent of symbiotic nodule formation. Symbiotic nitrogen-fixing bacteria and leguminous plants have evolved complex signal exchange mechanisms that enable specific bacterial species to induce specific host plant species to form invasion structures through which the bacterium enters the plant root. Once inside, the bacteria are enclosed within a micro-aerobic cell compartment by the host plant and differentiate into specialized cells that express an oxygen-sensitive enzyme to catalyze the conversion of nitrogen to ammonia for use by the host plant. Although the rhizobia-legume microbe-host relationship is mutually beneficial, rhizobial infections of plants can be deleterious. Rhizobia are similar to Brucella spp. that infect animals (including humans) in that they both form chronic infections of eukaryotic cells within a host-derived membrane compartment and require host-derived factors for survival within the host. Based on recent reports of increased virulence of Salmonella typhimurium, Pseudomonas aeruginosa, and Streptococcus pneumoniae in reduced gravity, differences in the infectious and/or pathogenic responses of rhizobial bacteria or the systemic acquired resistance of the host plant are also likely. [Ed. note: 1/19/2012] The Principal Investigator (Gary Stutte, PhD) and co-Investigator (Mike Roberts, PhD) had a change of employment from Dynamac Corporation, prime contractor of the Life Science Services Contract (LSSC), at Kennedy Space Center to QinetiQ-North America, the prime contractor for the Engineering Services Contract (ESC) at Kennedy Space Center effective 1 March, 2011.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The SyNRGE experiment investigates the role of microgravity in development of biological nitrogen fixation in legume species. This symbiosis between a legume and a nitrogen fixing bacteria is directly and indirectly responsible for over 20% of the protein in human diets. SyNRGE results are investigating underlying mechanisms of this symbiosis that has wide ranging agricultural and environmental applications.
Task Progress:	Two sets of experiments were conducted in orbit and in 24-hour delayed ground controls. The first was designed to determine if S. meliloti infect M. truncatula and initiate physiological changes associated with nodule formation. The second was designed to determine if microgravity altered the process of bacterial infection and host plant nodule formation. These experiments demonstrated that S. meliloti was able to infect the roots of M. truncatula and induce the production of nodulation factors that trigger early events of biological nitrogen fixation. SyNRGE also provided evidence that exposure to microgravity conditions in the PDFUs reduced the ability of the M. truncatula to form nodules upon return to 1g. This is the first evidence that the plant/microbe interaction necessary to initiate biological nitrogen fixation is achievable in microgravity, but suggests that the microgravity environment may have a negative impact upon development of functional nodules. Plant/Microbe interactions leading to biological nitrogen fixation This experiment was designed to determine if the nitrogen fixing bacteria, S. meliloti, infect M. truncatula and initiate physiological changes associated with nodule formation. Roots of five-day-old M. truncatula cultivar Jemalong A17 (Enod11::gus) seedlings were inoculated with either S. meliloti strain 1021 or strain ABS7 and integrated into BRIC-PDFU hardware before being placed in a 4°C Cold Bag for launch on Atlantis. Inoculated plants and uninoculated controls were in BRIC/PDFU hardware for 11 days, fixed in RNAlaterTM by the crew, and returned to Earth for analysis. M. truncatula cv. Jemalong A17 (Enod::gus) was fixed with RNAlaterTM on orbit, then stained using dyes to localize ß-glucuronase production assocation with activation of Enod 11 gene by S. meliloti. These test found that Enod11 had been activated in some of µg samples suggested that S. meliloti infected the root and that the Enod11::gus gene is activated under microgravity conditions. The site of bacterial infection was co-localized with the Enod11 activation site by staining for hemA::LacZ marker inserted into S. meliloti ABS7. The activation of Enod11, along with co-localization of S. meliloti, provide the first demonstration of bacterial induction of critical events in plant roots necessary to trigger formation of nodulation factors required to activate the genes necessary for nodule formation on the roots of legumes. Plant grown in PDFUs for 11 days exhibited etiolation and loss of chlorophyll, as would be expected with dark conditions. Growth analysis showed that M. truncatula seedlings grown in microgravity were 48% longer than ground controls (14.8 mm vs. 10 mm). The differential growth between µg and 1g was not observed with roots inoculated with S. meliloti. Microarray analysis (Affymetrix GeneChip Medicago Genome Array) samples identified 9,711genes that were differentially expressed (6967 up regulated/2744 down regulated) when comparing µg to 1g M. truncatula roots. Roots inoculated either S. meliloti strain had significantly fewer genes differentially expressed: 3,965 (1771 upregulated/2194 down-regulated) for Sm1012 and 3,778 (1411 up-regulated/2367 down-regulated) for SmABS7. This suggests that the detection of bacterial infection has signalled a physiological change in the carbon allocation pattern affecting root growth. Effect of Spaceflight conditions on nodulation potential of M. truncatula This experiment was designed to determine if microgravity altered the process of bacterial infection and host plant nodule formation. Seeds of two M. truncatula cultivars Jemalong A17 (Enod11::gus,) and SUNN, a super-nodulating mutant of A17, were germinated on orbit for 11 days in the middeck cabin and returned to Earth alive. S. meliloti strains Sm1021 and ABS7 were cultivated separately in broth culture on orbit in the fixative reservoir of the PDFU, and also returned to Earth alive. Upon landing, flight- and ground-control plants and bacteria were transferred from BRIC-PDFUs into Nunc™ 4-well plates for reciprocity crosses. Rates of plant growth and nodule development were measured over a period for 21 days The µg environment reduced the ability of both M. truncatula strains to form nodules upon return to Earth, relative to the 1g control plants. This response could be due to either a direct effect on the plant tissue and/or unique components of the unique growth environment of PDFU (darkness, elevated CO2, VOC accumulation, lack of convective mixing) experienced during STS-135. The reduction in number of plants which formed nodules was most prevalent in the

	super-nodulating line, SUNN, with 75% of the plants failing to produce nodules, in comparison to 50% of the ground controls.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Dissertations and Theses	Newsham G. "Germination of Medicago truncatula for Enclosed Enviroments." Thesis, Limerick Institute of Technology, Limerick, Ireland, December 2014. , Dec-2014
NASA Technical Documents	Stutte GW, Monje O, Wheeler RM. "ISS Research Guide: Plant Science." ISS Program Science Office, January 2015. , Jan-2015
Papers from Meeting Proceedings	Roberts MS, Stutte GW. "Microgravity effects on the early events of biological nitrogen fixation in Medicago truncatula: Results from the SyNRGE flight experiment." 1st Annual International Space Station (ISS) Research and Development Conference, Denver, CO, June 26-28, 2012. 1st Annual International Space Station (ISS) Research and Development Conference, Denver, CO, June 26-28, 2012. (AAI Science and Technology Series 114.20130000:111.) , Dec-2012